AgriSmart AI: Crop Recommendation & Precision Advisory System

College Engineering & Data Science Project - Comprehensive Report

1. Executive Summary

Selecting inappropriate crops or mismanaging chemical fertilizers degrades soil health, squanders capital, and drastically lowers crop yield. This platform applies supervised ensemble machine learning to analyze N, P, K soil macronutrients, pH, ambient temperature, humidity, and rainfall to prescribe high-probability crop matches and tailored fertilizer schedules.

2. AI Implementation Workflow

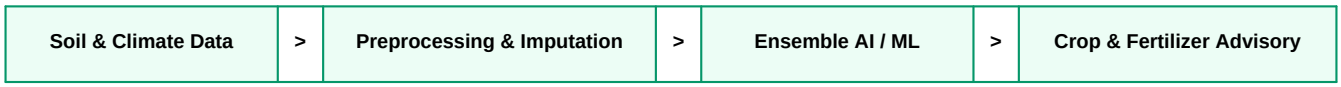


Fig 1: Machine Learning & Agronomic Advisory Pipeline

Stage	Implementation Details
Data	3840 records across 32 crops. Input features: N, P, K (kg/ha), temperature (°C), relative humidity (%), soil pH, and rainfall (mm).
Processing	Deduplication, percentile clipping, median imputation of 161 missing values, standard scaling, and stratified split (3072 train / 768 test).
AI / ML	Benchmarked Random Forest, Extra Trees, KNN, and Gaussian Naive Bayes using 5-fold stratified cross-validation.
Output	Multi-class probabilistic ranking, macronutrient deficit calculator, pH neutralization recommendations, and exportable farm advisory.

3. Model Benchmark & Results

Model Candidate	5-Fold CV Accuracy	Held-out Test Accuracy
Random Forest	94.69%	94.01%
Extra Trees	95.31%	94.14%
K-Nearest Neighbors	90.17%	89.97%
Naive Bayes	95.54%	94.66%

Optimal Classifier: **Naive Bayes** with **94.66%** accuracy.

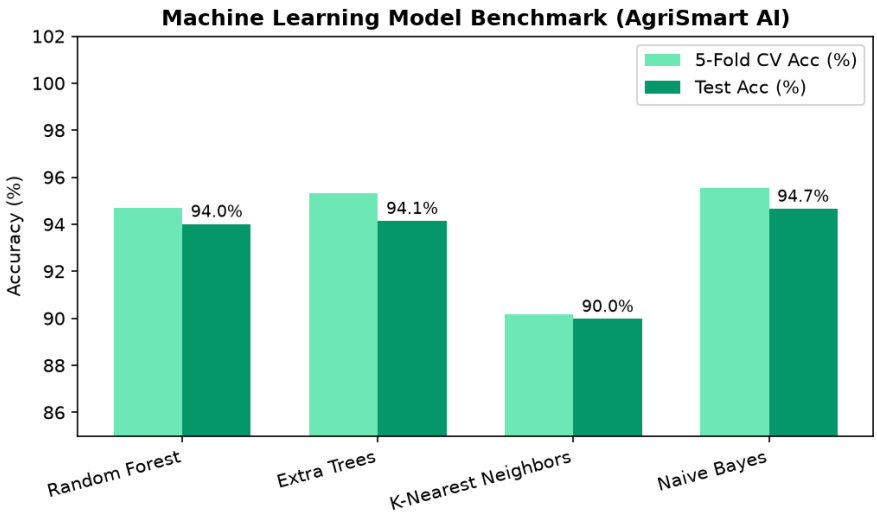


Fig 2: Model Comparison (Cross-Validation vs Test Set)

4. Agronomic Feature Importance

Feature importance analysis identifies which environmental and soil parameters contribute most strongly to crop classification decisions.

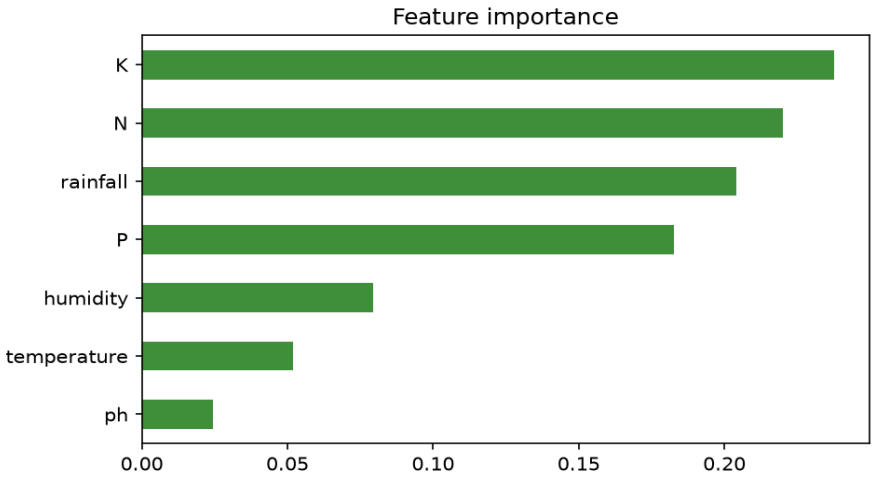


Fig 3: Feature Importance Analysis (Gini Impurity Metric)

5. Confusion Matrix

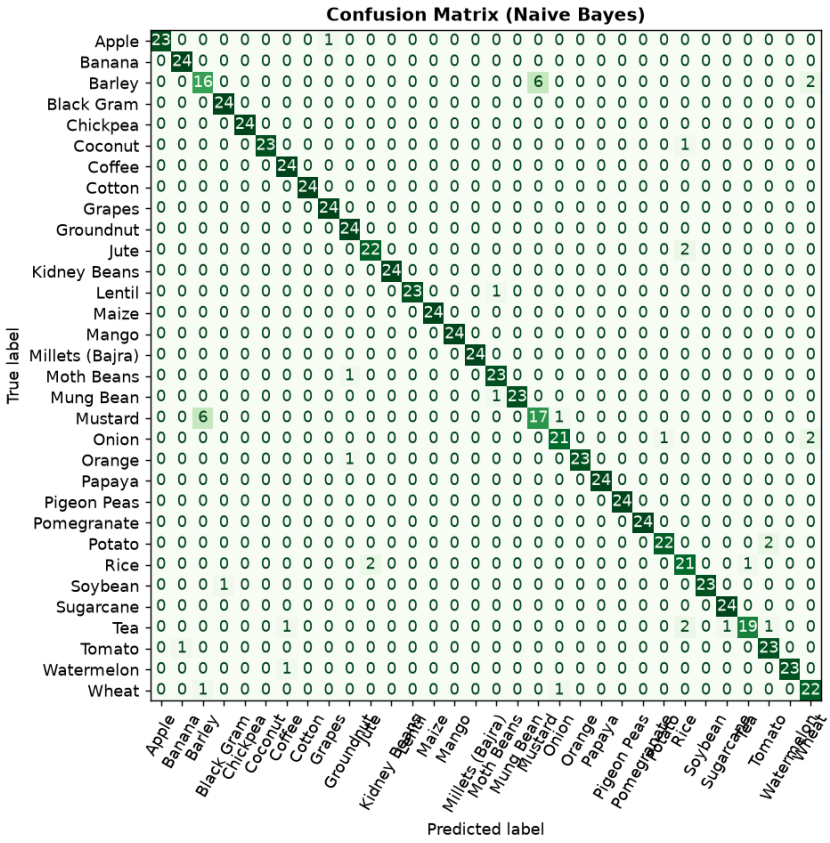


Fig 4: Multi-Class Confusion Matrix

6. Real-World Field Evaluation & Recommendations

Field Scenario	Top Ranked Crops	Precision Soil Advisory
Field 1 - Punjab Indo-Gangetic Alluvial	Wheat (99%), Onion (1%), Barley (0%)	Soil N-P-K nutrient status is in near-optimal balance for Wheat.
Field 2 - Maharashtra Black Cotton Soil	Cotton (100%), Maize (0%), Banana (0%)	Soil N-P-K nutrient status is in near-optimal balance for Cotton.
Field 3 - Coastal Karnataka Horticultural Land	Banana (100%), Tomato (0%), Jute (0%)	Soil N-P-K nutrient status is in near-optimal balance for Banana.
Field 4 - Arid Rajasthan Sandy Soil	Millets (Bajra) (100%), Black Gram (0%), Mustard (0%)	Soil N-P-K nutrient status is in near-optimal balance for Millets (Bajra).

7. Deployment & Real-World Utility for Farmers

The upgraded AgriSmart AI system bridges theoretical machine learning with actionable farm management. By incorporating beginner farmer soil wizards, crop life-cycle roadmaps, economic ROI estimation, live weather spray advisories, and multi-lingual voice accessibility, the application addresses the actual challenges faced by both novice and experienced agriculturists.